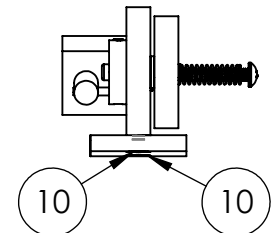
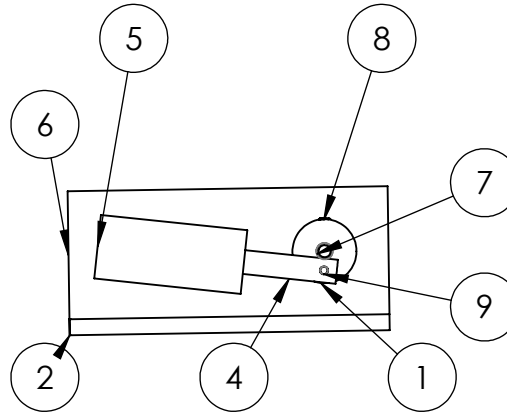
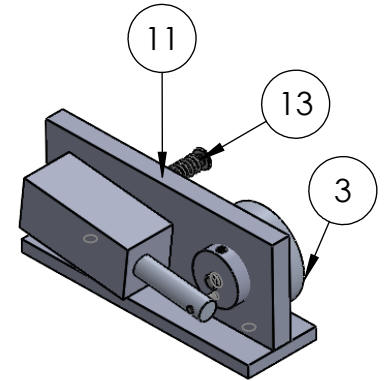
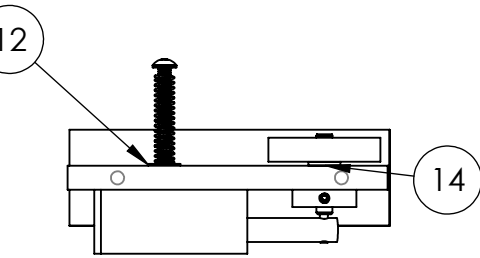


B

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Air Engine Crank		1
2	Air Engine Baseplate		1
3	Air Engine Flywheel		1
4	Air Engine Piston		1
5	Air Engine Cylinder		1
6	Air Engine Frame		1
7	98381A544	Axle-1/4" Dowel Pin 1" Long	1
8	92313A192	8-32 x 3/8" Set Screw	1
9	98381A473	Crank-1/8" Dowel Pin 3/4" Long	1
10	92196A267	10-32 x 3/8" SHCS Frame Screw	2
11	90272A837	10-32 x 1 3/4" PHPS Pivot Screw	1
12	95606A431	Washer #10	1
13	9657K35	Spring 1.375" Long 0.312 OD	1
14	6338K412	Bronze Bushing 1 1/4	1



A

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL ±
ANGULAR: MACH ± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

INTERPRET GEOMETRIC
TOLERANCING PER:

MATERIAL

FINISH

DO NOT SCALE DRAWING

DRAWN	Emerson Johnson	NAME	DATE
CHECKED			
ENG APPR.			
MFG APPR.			
Q.A.			
COMMENTS:			

TITLE:
Air Engine Assembly

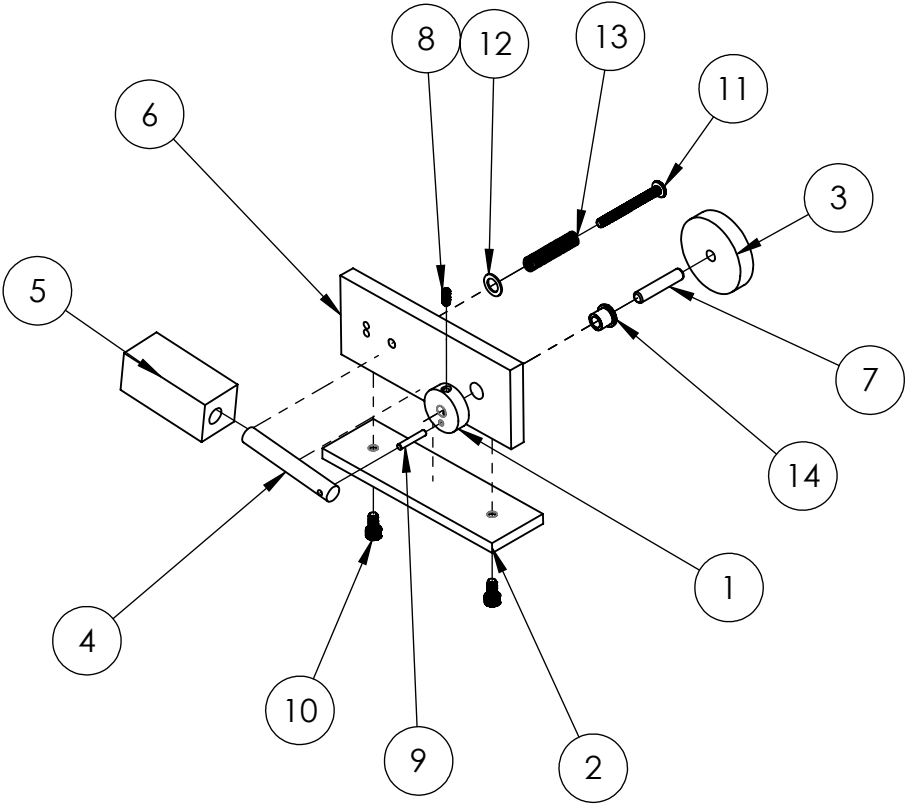
SIZE	DWG. NO.	REV
A	1	
SCALE: 1:3	WEIGHT:	SHEET 1 OF 1

B

A

B

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Air Engine Crank		1
2	Air Engine Baseplate		1
3	Air Engine Flywheel		1
4	Air Engine Piston		1
5	Air Engine Cylinder		1
6	Air Engine Frame		1
7	98381A544	Axle -1/4 "DowelPin 1' long	1
8	92313A192	8-32×78 Set Screw	1
9	98381A473	Crank - 1/8" DowelPin 3/4 long	1
10	92196A267	10-32 x 3/8 " SHCS Frame Screw	2
11	90272A837	10-32 x 1 3/4 " Phps Pivot Screw	1
12	95606A431	Washer #10	1
13	9657K35	Spring 1.375 " long 0.31200	1
14	6338K412	Bronze Bushing 1 1/4 "	1



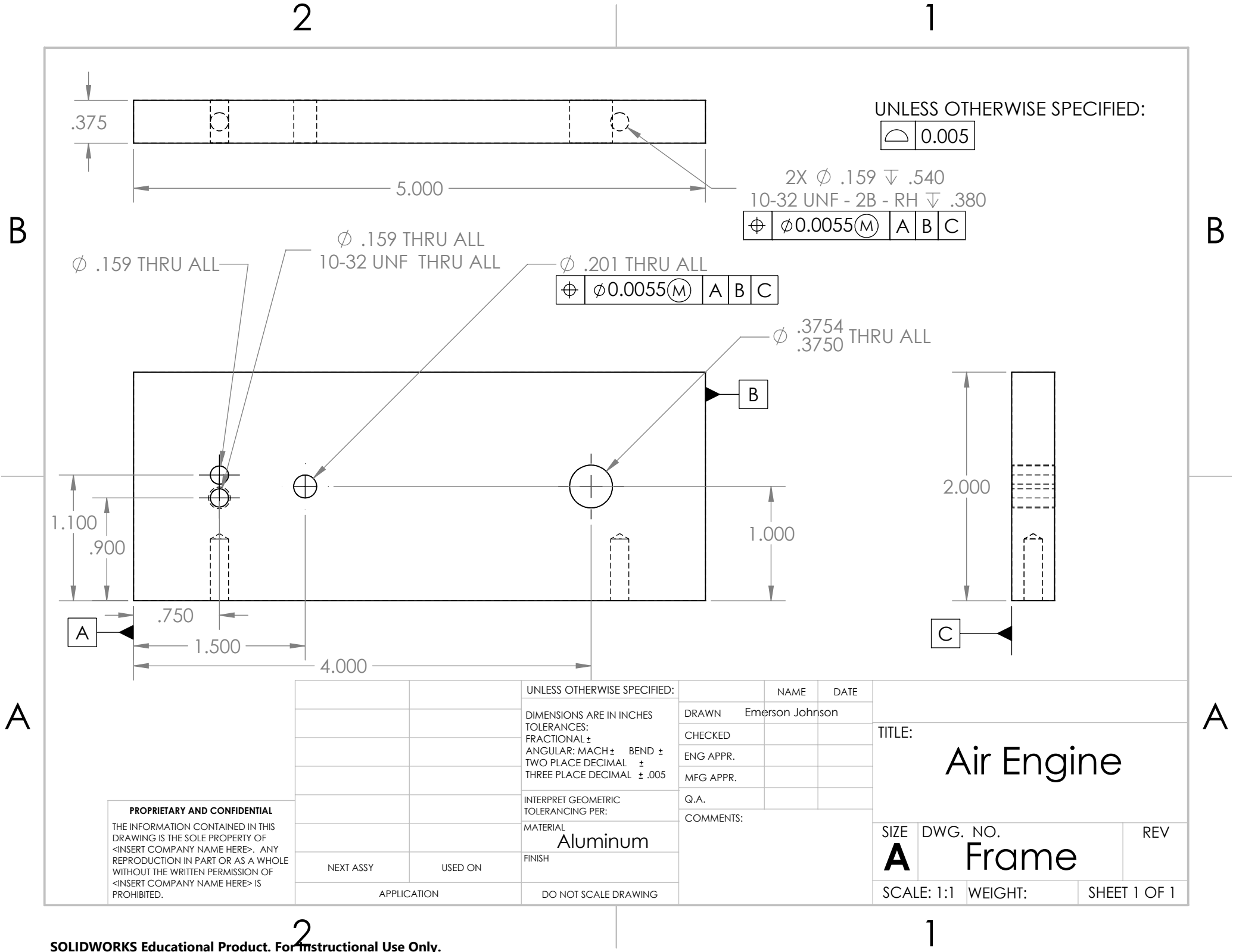
B

A

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.			UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: Air Engine
			DIMENSIONS ARE IN INCHES	DRAWN	Emad Alomeri		
			TOLERANCES:	CHECKED			
			FRACTIONAL ±	ENG APPR.			
			ANGULAR: MACH ± BEND ±	MFG APPR.			
			TWO PLACE DECIMAL ±	Q.A.			
		THREE PLACE DECIMAL ±	COMMENTS:				
		INTERPRET GEOMETRIC TOLERANCING PER:					
		MATERIAL					
		Aluminum					
		FINISH					
	NEXT ASSY	USED ON					
	APPLICATION		DO NOT SCALE DRAWING				

SIZE	DWG. NO.	REV
A	Exploded View	
SCALE: 1:5	WEIGHT:	SHEET 1 OF 1

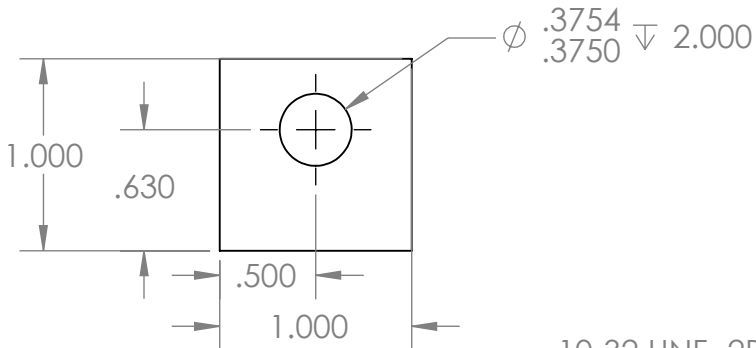
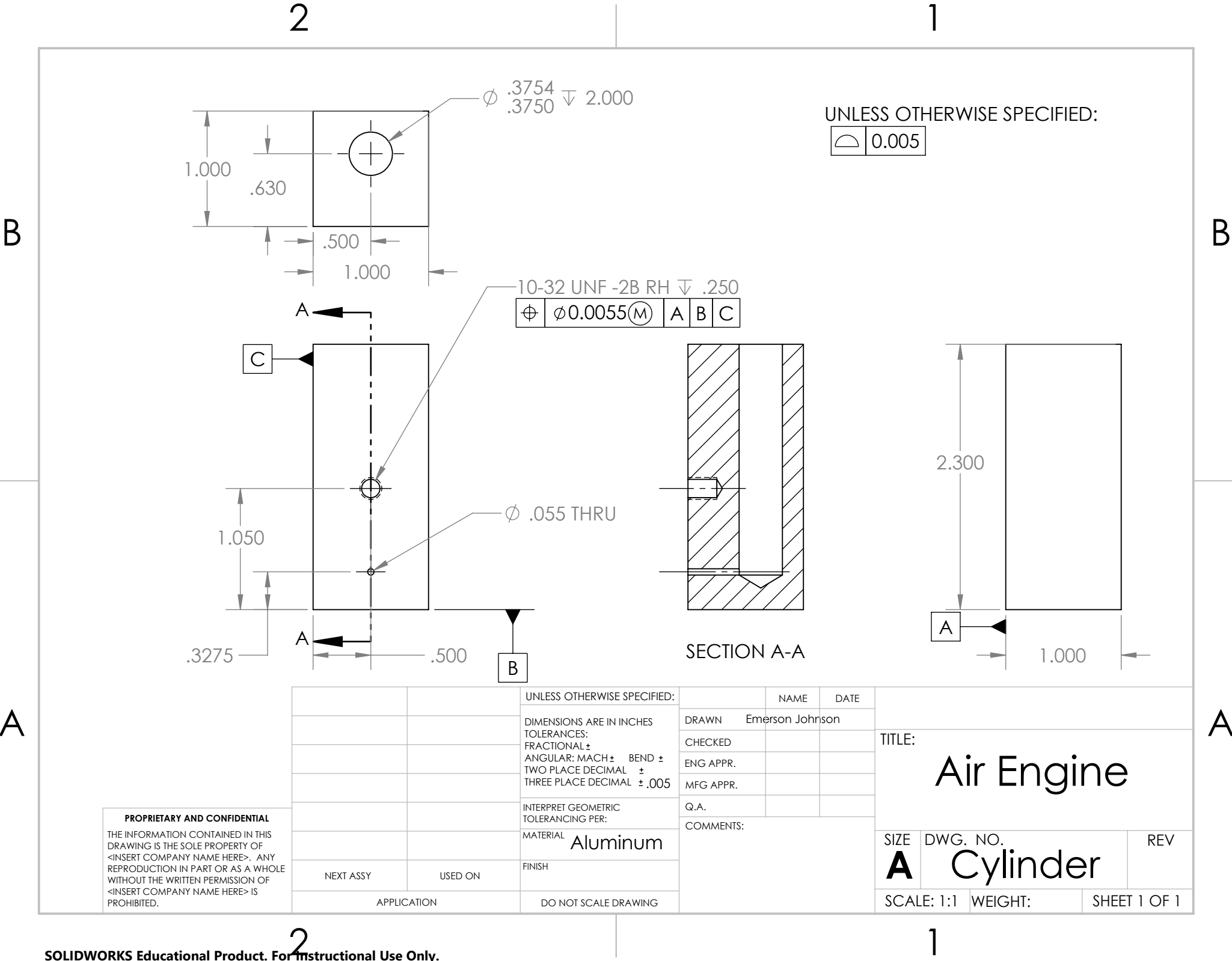
A



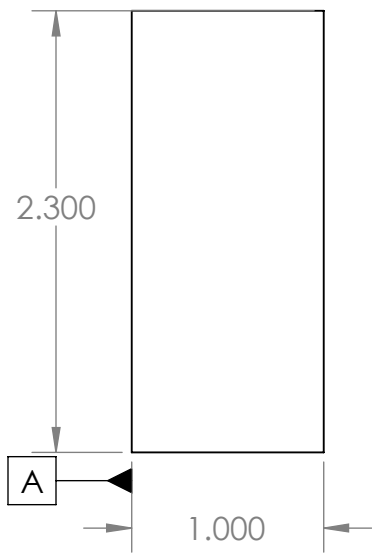
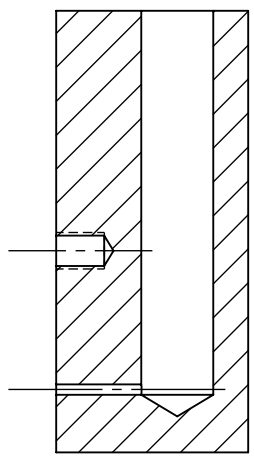
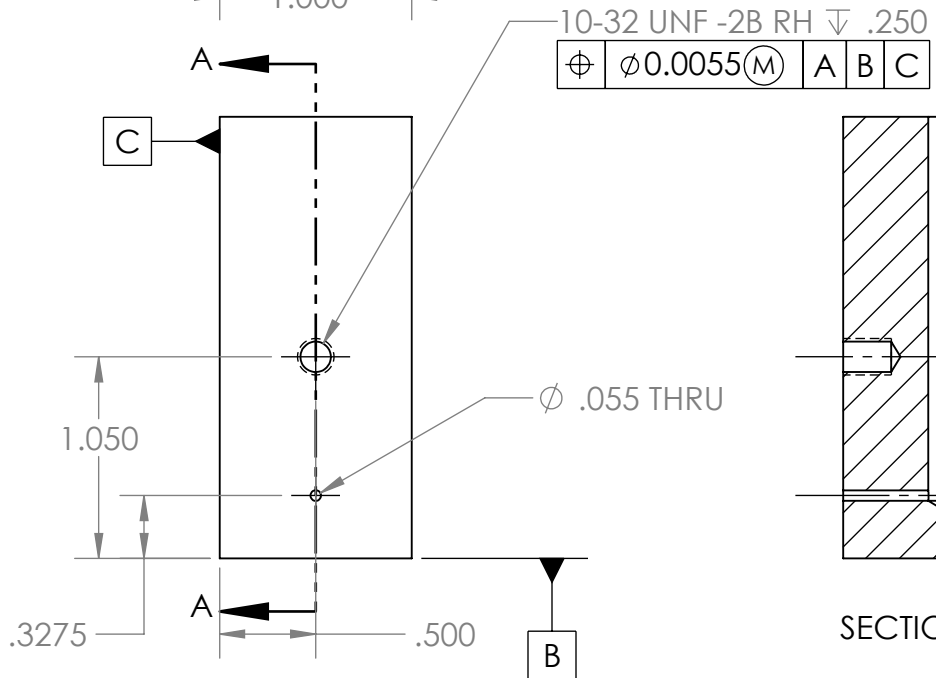
PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL ±
		ANGULAR: MACH ± BEND ±
		TWO PLACE DECIMAL ±
		THREE PLACE DECIMAL ± .005
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL
		Aluminum
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

TITLE:			
Air Engine			
SIZE	DWG. NO.	REV	
A	Frame		
SCALE: 1:1	WEIGHT:	SHEET 1 OF 1	



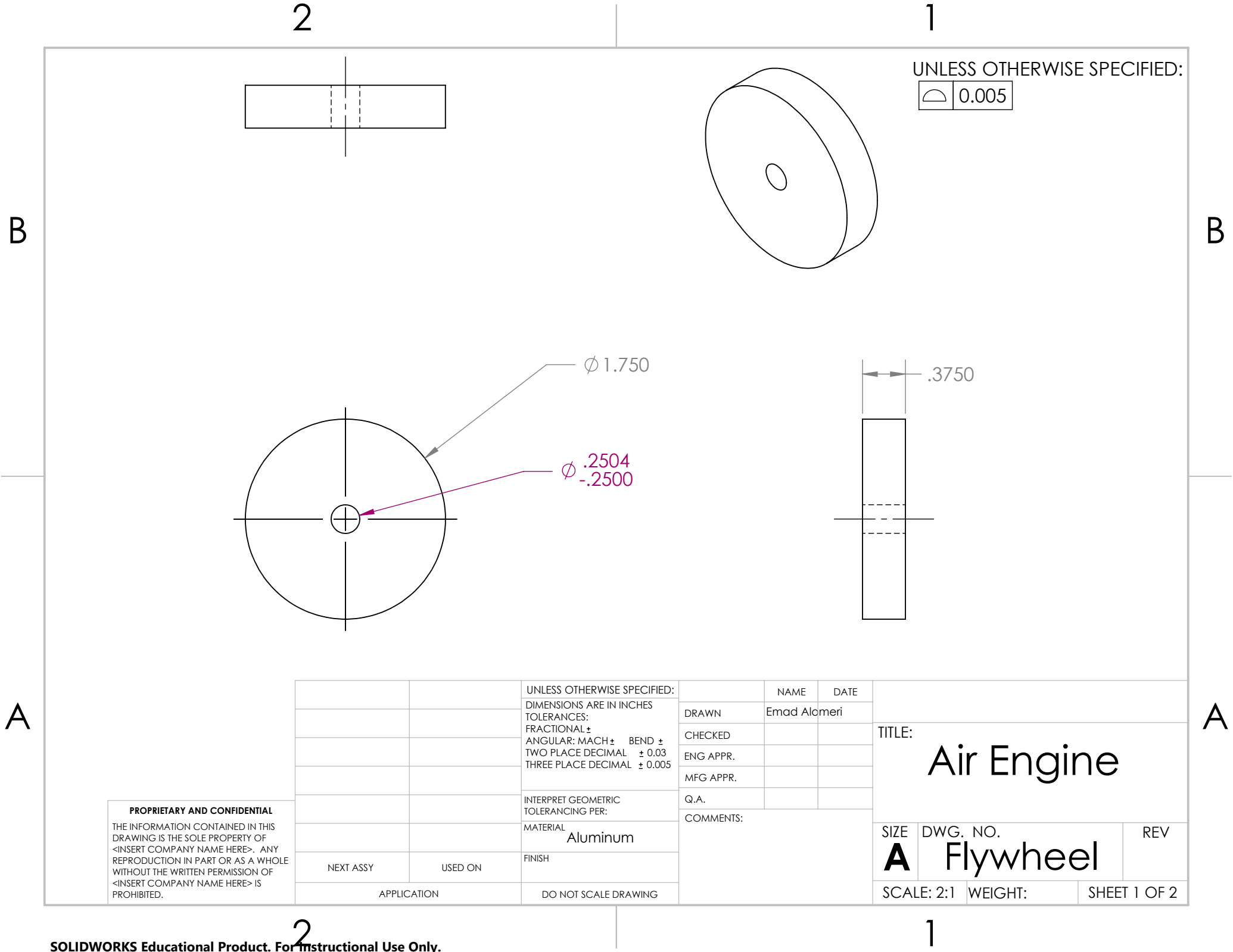
UNLESS OTHERWISE SPECIFIED:
 ∇ 0.005



PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES	DRAWN	Emerson Johnson	
		TOLERANCES:	CHECKED		
		FRACTIONAL $\pm$	ENG APPR.		
		ANGULAR: MACH $\pm$ BEND $\pm$	MFG APPR.		
		TWO PLACE DECIMAL $\pm$	Q.A.		
		THREE PLACE DECIMAL $\pm .005$	COMMENTS:		
		INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL Aluminum			
		FINISH			
NEXT ASSY	USED ON				
APPLICATION		DO NOT SCALE DRAWING			

TITLE: Air Engine			
SIZE A	DWG. NO. Cylinder	REV	
SCALE: 1:1	WEIGHT:	SHEET 1 OF 1	



A

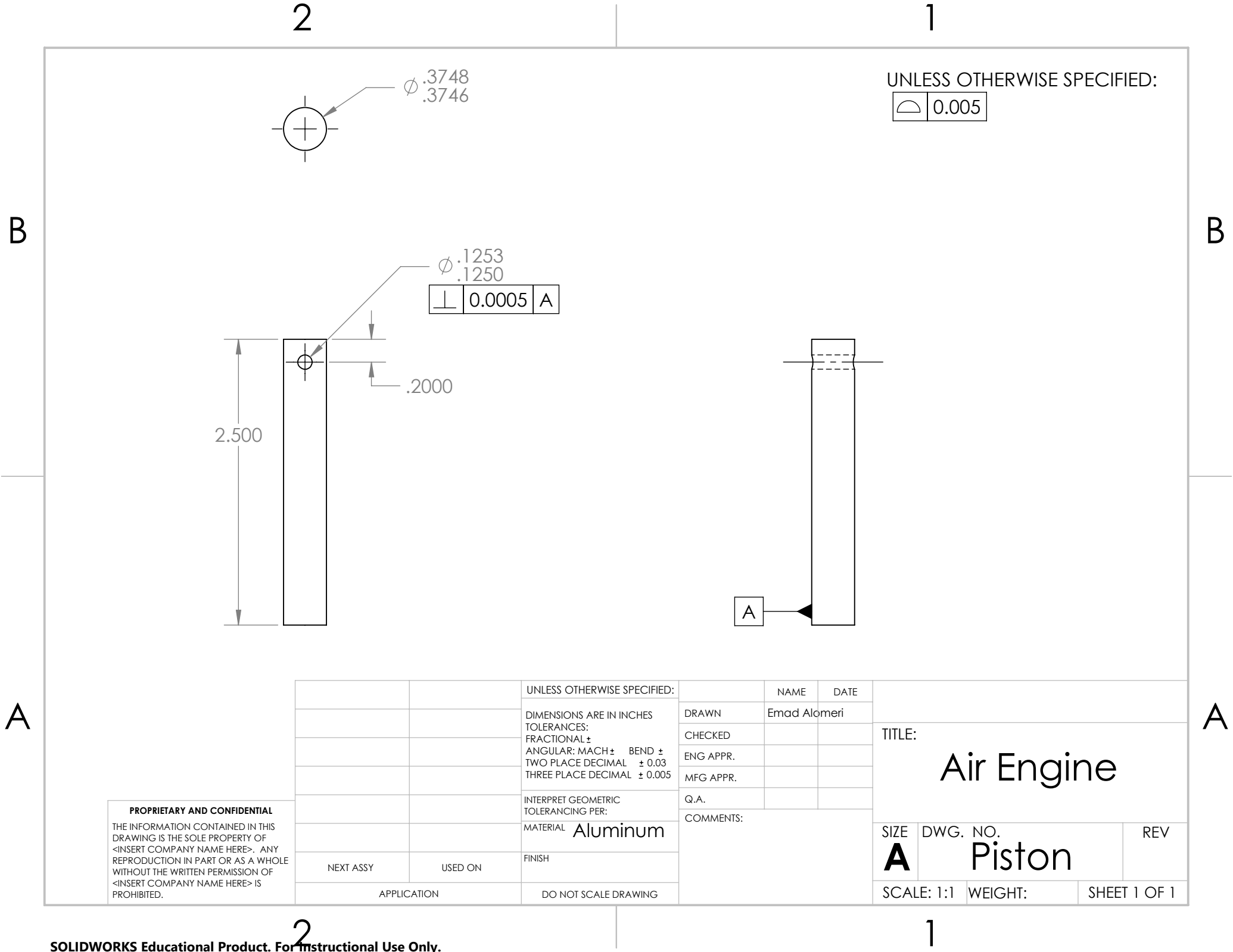
A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL $\pm$ ANGULAR: MACH $\pm$ BEND $\pm$ TWO PLACE DECIMAL ± 0.03 THREE PLACE DECIMAL ± 0.005
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL Aluminum
NEXT ASSY	USED ON	FINISH
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	Emad Alomeri	
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE: Air Engine		
SIZE A	DWG. NO. Flywheel	REV
SCALE: 2:1	WEIGHT:	SHEET 1 OF 2



UNLESS OTHERWISE SPECIFIED:

	0.005
--	-------

ϕ .3748
 ϕ .3746

ϕ .1253
 ϕ .1250

	0.0005	A
--	--------	---

.2000

2.500

A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS
DRAWING IS THE SOLE PROPERTY OF
<INSERT COMPANY NAME HERE>. ANY
REPRODUCTION IN PART OR AS A WHOLE
WITHOUT THE WRITTEN PERMISSION OF
<INSERT COMPANY NAME HERE> IS
PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES	DRAWN	Emad Alomeri	
		TOLERANCES:	CHECKED		
		FRACTIONAL $\pm$	ENG APPR.		
		ANGULAR: MACH $\pm$ BEND $\pm$	MFG APPR.		
		TWO PLACE DECIMAL $\pm$ 0.03	Q.A.		
		THREE PLACE DECIMAL $\pm$ 0.005	COMMENTS:		
		INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL Aluminum			
NEXT ASSY	USED ON	FINISH			
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

Air Engine

SIZE

A

DWG. NO.

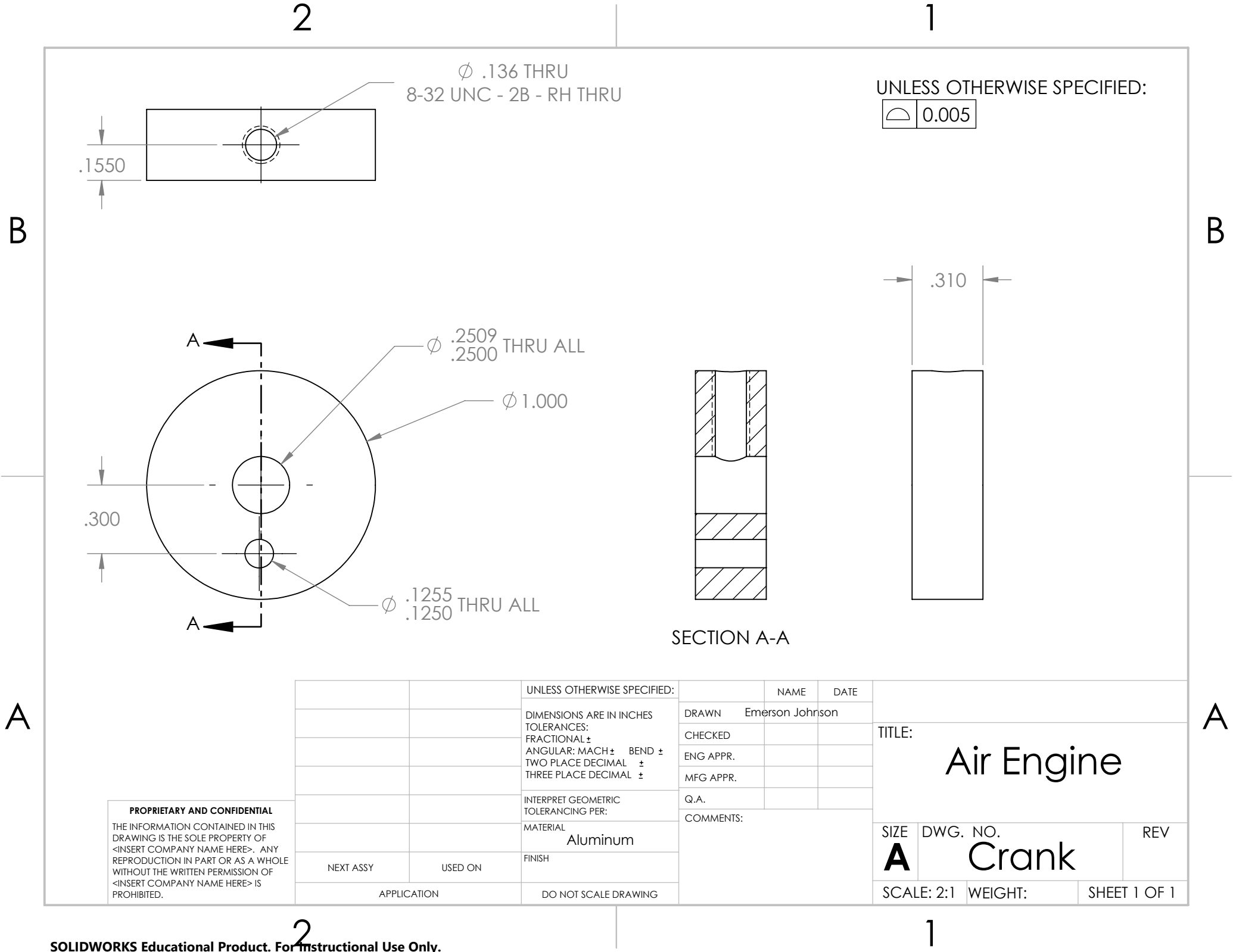
Piston

REV

SCALE: 1:1

WEIGHT:

SHEET 1 OF 1



UNLESS OTHERWISE SPECIFIED:

	0.005
--	-------

SECTION A-A

A

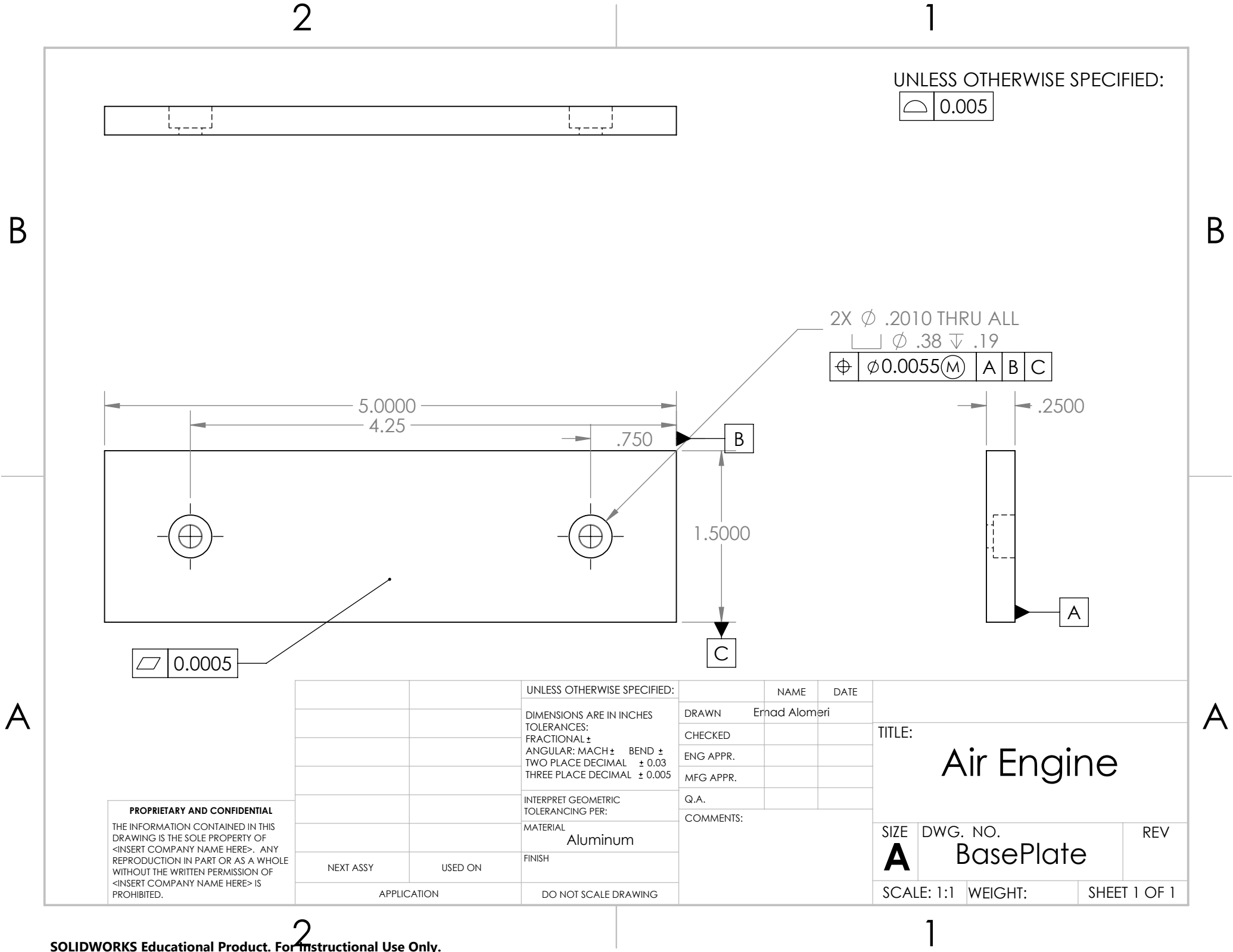
A

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL $\pm$
		ANGULAR: MACH $\pm$ BEND $\pm$
		TWO PLACE DECIMAL $\pm$
		THREE PLACE DECIMAL $\pm$
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL
		Aluminum
		FINISH
NEXT ASSY	USED ON	
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	Emerson Johnson	
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

TITLE:		
Air Engine		
SIZE	DWG. NO.	REV
A	Crank	
SCALE: 2:1	WEIGHT:	SHEET 1 OF 1



B

B

A

A

UNLESS OTHERWISE SPECIFIED:

0.005

2X Ø .2010 THRU ALL

Ø .38 ± .19

Ø 0.0055 (M) A B C

0.0005

PROPRIETARY AND CONFIDENTIAL
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:
		DIMENSIONS ARE IN INCHES
		TOLERANCES:
		FRACTIONAL ±
		ANGULAR: MACH ± BEND ±
		TWO PLACE DECIMAL ± 0.03
		THREE PLACE DECIMAL ± 0.005
		INTERPRET GEOMETRIC TOLERANCING PER:
		MATERIAL
		Aluminum
		FINISH
NEXT ASSY	USED ON	
APPLICATION		DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	Emad Alomeri	
CHECKED		
ENG APPR.		
MFG APPR.		
Q.A.		
COMMENTS:		

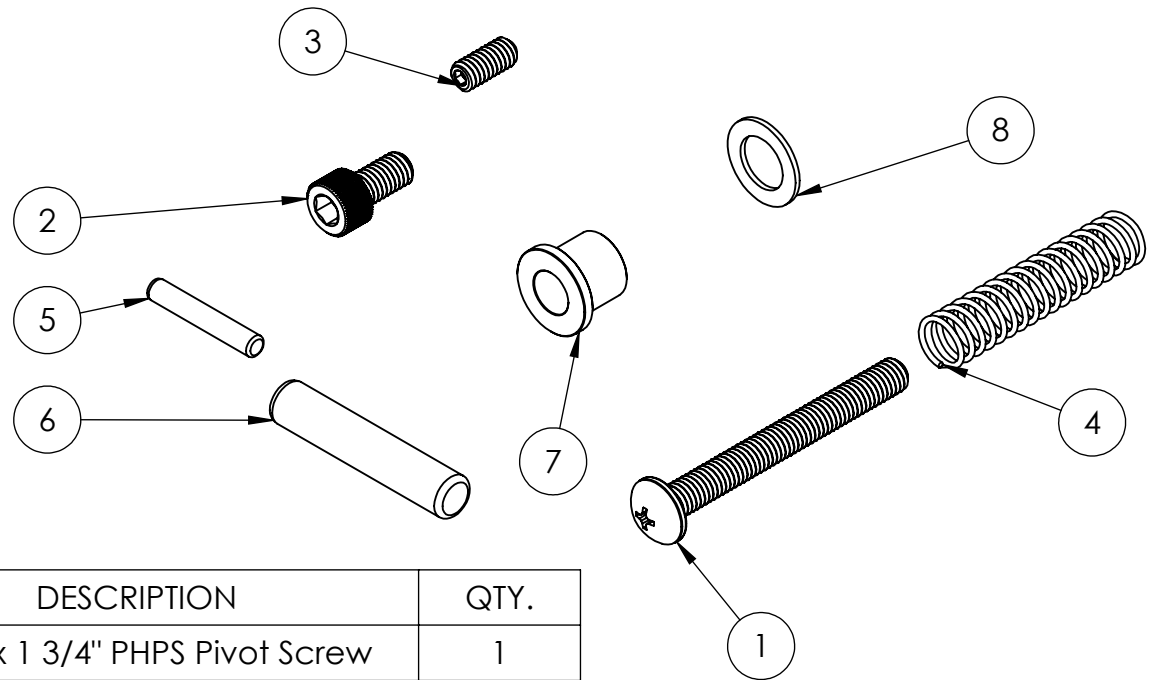
TITLE:

Air Engine

SIZE	DWG. NO.	REV
A	BasePlate	
SCALE: 1:1	WEIGHT:	SHEET 1 OF 1

B

B



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	90272A837	10-32 x 1 3/4" PHPS Pivot Screw	1
2	92196A267	10-32 x 3/8" SHCS Frame Screw	1
3	92313A192	8-32 x 3/8" Set Screw	1
4	9657K35	Spring 1.375" Long 0.312 OD	1
5	98381A473	Crank - 1/8" Dowel Pin 3/4" Long	1
6	98381A544	Axle - 1/4" Dowel Pin 1" Long	1
7	6338K412	Bronze Bushing 1 1/4	1
8	95606A431	Washer #10	1

A

A

PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.			UNLESS OTHERWISE SPECIFIED:	NAME	DATE	TITLE: Purchased Parts
			DIMENSIONS ARE IN INCHES	DRAWN	Emerson Johnson	
			TOLERANCES:	CHECKED		
			FRACTIONAL ±	ENG APPR.		
			ANGULAR: MACH ± BEND ±	MFG APPR.		
			TWO PLACE DECIMAL ±	Q.A.		SIZE DWG. NO. REV A 9
			THREE PLACE DECIMAL ±	COMMENTS:		
			INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL				
		FINISH				
	NEXT ASSY	USED ON				
	APPLICATION		DO NOT SCALE DRAWING			SCALE: 1:1 WEIGHT: SHEET 1 OF 1